

A Prospective Randomized Comparison using MRI of Two Distinct Allogenic Tissue Constructs for Anterior Cruciate Ligament Reconstruction

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Objectives: To compare the radiographic and clinical outcomes of anterior cruciate ligament (ACL) reconstruction using quadrupled hamstring tendon (HT) allograft vs. doubled tibialis anterior (TA) allograft.

Methods: Methods: A prospective randomized controlled trial (PRCT) at a single center was conducted with enrollment from 7/2010 to 4/2012. One hundred subjects undergoing primary ACL reconstruction were randomized to either HT or TA allograft (47 vs. 53, respectively). Subjects completed KOOS, IKDC, Tegner, VR-12, and Lysholm outcomes measure pre-operatively, and then post-operatively at the 6 month and 24 month time period. Arthrometric testing using KT-1000 (MEDmetric® Corporation) was applied at 6 months post-operatively. At this time patients also underwent MRI to evaluate allograft integrity and ligamentization. The signal to noise quotient (SNQ) was calculated via the Impax markup tool (Afga HealthCare®) using sagittal non-contrast T2 weighted MRI (Figure 1) as follows: $SNQ = (\text{Signal graft} - \text{Signal Quadriceps}) / \text{Signal Background}$

Results: MRIs were obtained in the HT (n=22, 53%, 205.6 ± 35.7 days) and TA (n=23, 47%, 187.3 ± 26.0 days) groups. The average SNQ for the HT and TA allografts were 2.94 ± 3.03 and 4.14 ± 3.64 , respectively ($p = 0.12$). There was no correlation between MRI SNQ and subject age, BMI, sex, graft diameter or smoking status. With arthrometric testing there was a trend for the HT allografts to have increased laxity as compared to contralateral side but the magnitude of difference was not clinically relevant (average 0.88 vs. 0.40mm, respectively). At an average of 2.4 ± 0.4 years there was no difference detected between groups in any of the outcomes scores with each showing improvement from baseline (Table 1). Allograft re-tear rates were similar between groups (9.5% HT vs. 4.5% TA, $p=1.0$).

Conclusion: Limited data exists directly comparing different ACL allograft soft tissue constructs. Specifically, to the author's knowledge this is the first prospective RCT comparing quadrupled hamstring tendon and doubled tibialis anterior allografts. Our data indicate no difference in graft failure rate, rate of ligamentization, and similar significant improvement from baseline with respect to subjective outcomes scores between each treatment group after more than two years.

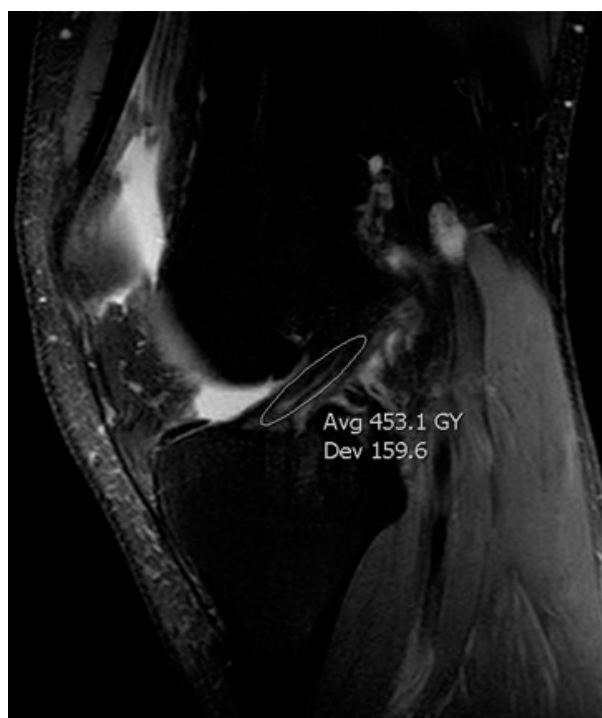


Figure 1. Representative MRI of reconstructed ACL. A standard ellipse demarcating the intra-articular portion of the graft was defined for analysis. The sagittal non-contrast T2 weighted MRI cut containing the largest portion of the reconstructed ACL was used for analysis. The signal to noise quotient (SNQ) was calculated for each graft.

Patient Reported Outcomes	Pre-Operative		24 Months Post-Op		Change from Baseline	
	HT	TA	HT	TA	HT	TA
IKDC ¹	45.9 ± 18.6	43.0 ± 14.5	76.7 ± 20.5	81.1 ± 12.6	27.4 ± 23.4*	38.2 ± 21.4*
VR-12 Physical ²	38.0 ± 9.6	35.7 ± 11.6	50.3 ± 8.5	53.1 ± 4.1	10.8 ± 10.6*	20.8 ± 17.0*†
KOOS ³	53.0 ± 19.1	49.4 ± 16.8	76.0 ± 21.3	81.2 ± 9.9	22.2 ± 20.0*	30.9 ± 19.2*
Lysholm ⁴	63.0 ± 20.2	56.1 ± 18.3	79.5 ± 20.2	86.0 ± 13.0	15.8 ± 25.4*	28.0 ± 22.2*
Tegner ⁵	2.6 ± 1.7	2.6 ± 1.3	5.7 ± 1.9	5.0 ± 1.7	3.2 ± 2.2*	3.5 ± 1.8*

Table 1. Patient Reported Outcomes. All values represent mean score ± standard deviation. *Statistically significant change from baseline, $p < 0.05$. † Statistically significant difference in change compared to HT group. 1. International Knee Documentation Committee Subjective Knee Score, score of 100 represents no limitation in activities of daily living. 2. Veteran Rand 12 Item Health Survey, Physical Health component, lower score represent decreased quality of life. 3. Knee Injury and Osteoarthritis Outcomes Score, score of 0 represents extreme knee problems and 100 represents no problems. 4. Lysholm Knee Score, score of 93-100 represent good to excellent ability to manage in everyday life; less than 65 represents poor ability. 5. Tegner Activity Score with Level 10 being a professional athlete and level 0 being disabled from work.